

GLOBAL
EDITION



Statistics for Managers Using Microsoft® Excel®

NINTH EDITION

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Chapter 5

Discrete Probability Distributions

Objectives

In this chapter, you will learn:

- The properties of a probability distribution.
- How to calculate the expected value and variance of a probability distribution.
- How to calculate probabilities from binomial, Poisson and hypergeometric distributions.
- How to use these discrete probability distributions to solve business problems.

Definitions

A **random variable** is a function that assigns a number to each outcome of an experiment.

Alternatively, the **value** of a random variable is a numerical event.

Instead of talking about the coin flipping event as

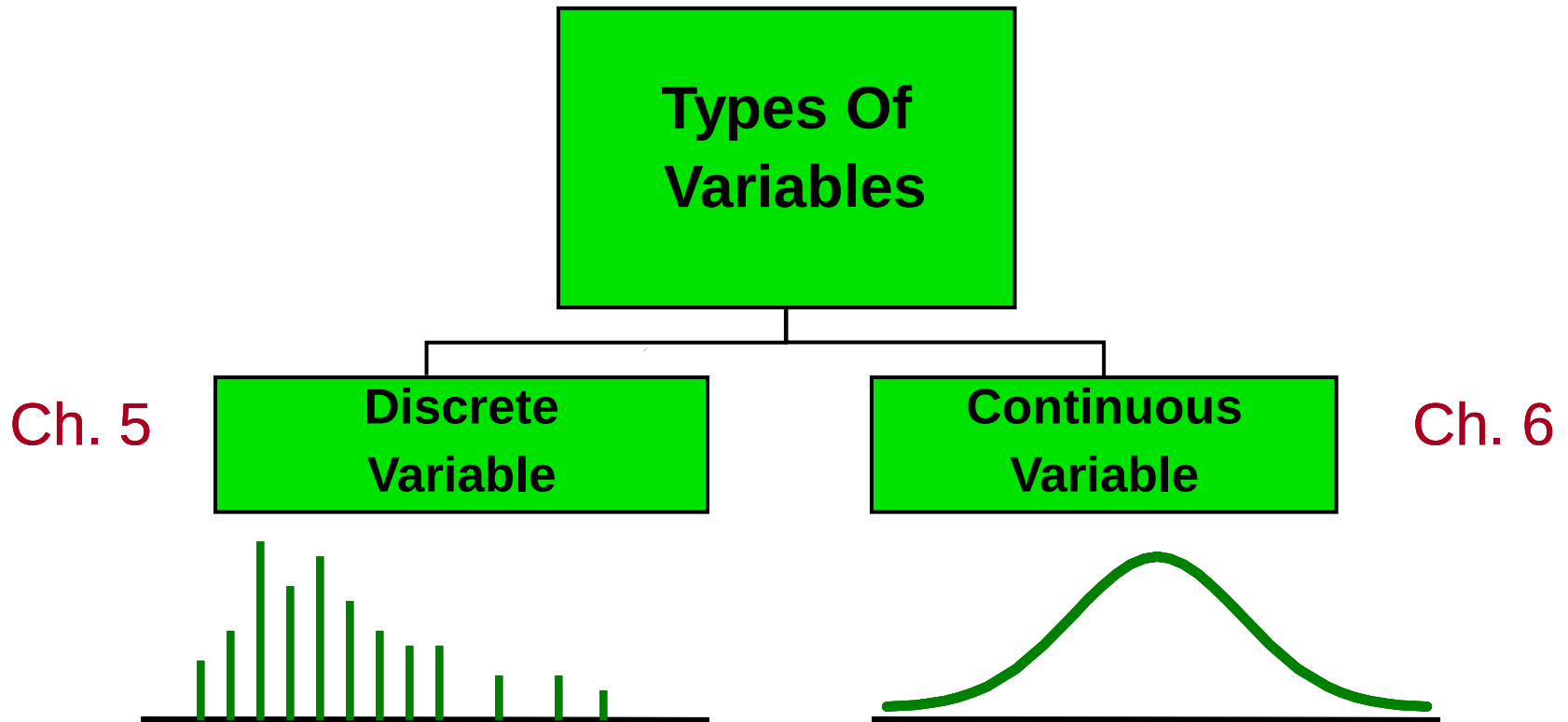
{heads, tails} think of it as


 $\{1, 0\}$ *“the number of heads when flipping a coin”*
 (numerical events)

Definitions

- **Discrete** variables produce outcomes that come from a counting process (e.g. number of classes you are taking).
- **Continuous** variables produce outcomes that come from a measurement (e.g. your annual salary, or your weight).

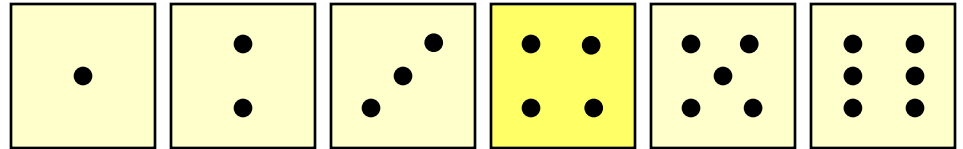
Types Of Variables



Discrete Variables

- Can only assume a countable number of values.

Examples:



- Roll a die twice

Let X be the number of times 4 occurs
(then X could be 0, 1, or 2 times).

- Toss a coin 5 times.

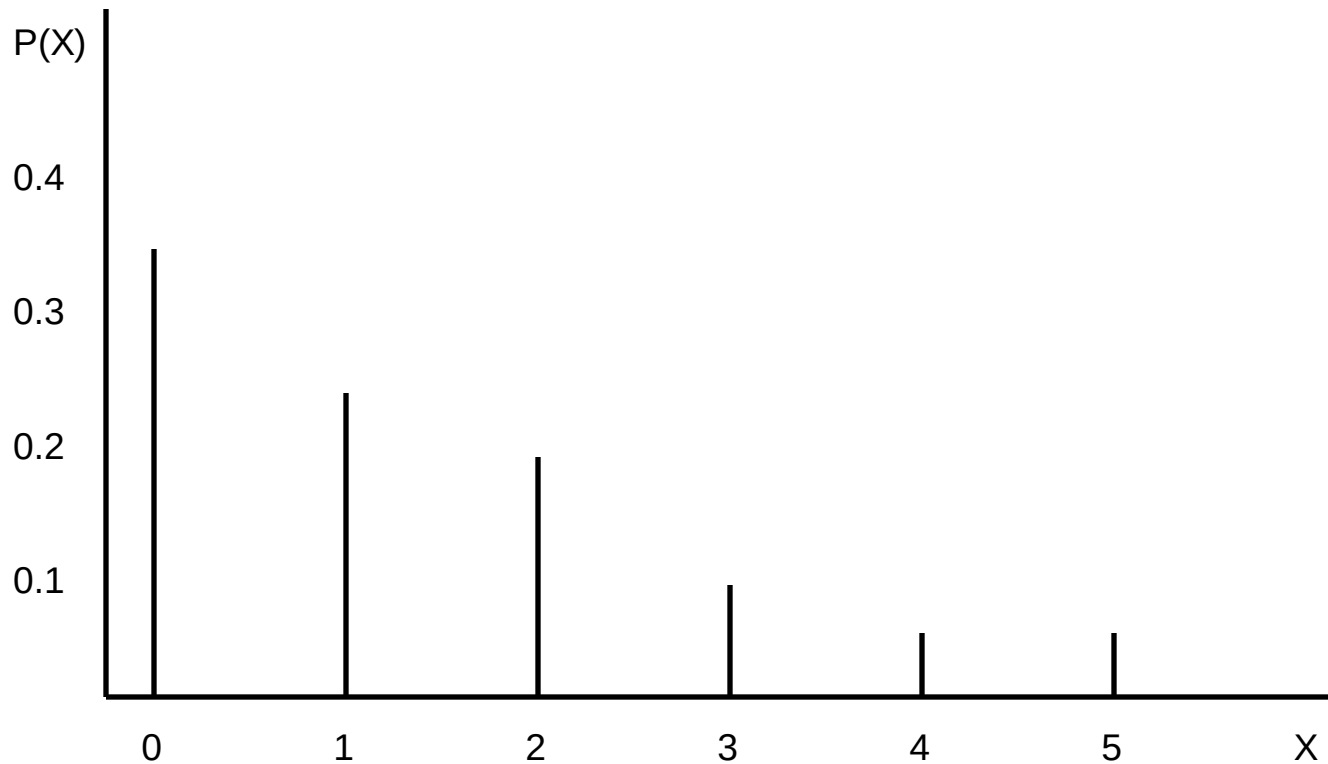
Let X be the number of heads
(then $X = 0, 1, 2, 3, 4, \text{ or } 5$).

Probability Distribution For A Discrete Variable

- A **probability distribution for a discrete variable** is a mutually exclusive list of all possible numerical outcomes for that variable and a probability of occurrence associated with each outcome.

Interruptions Per Day In Computer Network	Probability
0	0.35
1	0.25
2	0.20
3	0.10
4	0.05
5	0.05

Probability Distributions Are Often Represented Graphically



Probability Distributions Satisfy Two Conditions

The probabilities of the values of a discrete random variable may be derived by means of any applicable technique, the following two conditions must apply:

1. $0 \leq P(x) \leq 1$ for all x
2. $\sum_{all\ x} P(x) = 1$

Numerical Event (x_i)	Probability $P(X = x_i)$
0	0.35
1	0.5
2	0.15

Note:

1. $0 \leq P(x) \leq 1$

2. $\sum_{i=1}^3 P(x_i)$
 $= 0.35 + 0.5 + 0.15 = 1$

Expected Value Of Discrete Variables, Measuring Center

- **Expected Value (or mean)** of a discrete variable (Weighted Average):

$$\mu = E(X) = \sum_{i=1}^N x_i P(X = x_i)$$

Interruptions Per Day In Computer Network (x_i)	Probability $P(X = x_i)$	$x_i P(X = x_i)$
0	0.35	$(0)(0.35) = 0.00$
1	0.25	$(1)(0.25) = 0.25$
2	0.20	$(2)(0.20) = 0.40$
3	0.10	$(3)(0.10) = 0.30$
4	0.05	$(4)(0.05) = 0.20$
5	0.05	$(5)(0.05) = 0.25$
	1.00	$\mu = E(X) = 1.40$

Rules of Expectations

- Rule 1: $E[c] = c$
- Rule 2: $E[X + c] = E[X] + c$
- Rule 3: $E[c \cdot X] = c \cdot E[X]$

It therefore follows that, for constants a and b

$$E[aX + b] = a \cdot E[X] + b$$

Rules of Expectations:

Example

Monthly sales have a mean of R25 000 and a standard deviation of R4 000. Profits are calculated by multiplying sales by 30% and subtracting fixed costs of R6 000.

*Calculate the **MEAN** monthly profit.*

Describe the problem in algebraic terms:

- *Sales have a mean of R25 000* $\rightarrow E(\text{Sales}) = 25\ 000$
- *Profits are calculated by* $\rightarrow \text{Profit} = 0.30(\text{Sales}) - 6\ 000$

Rules of Expectations:

Example

Let X = the monthly sales. It is given that $E[X] = 25000$.

It is given that profit = $P = 0.3X - 6000$.

Hence,

$$\begin{aligned} E[0.3X + (-6000)] &= 0.3E[X] + (-6000) \\ &= 0.3E[X] - 6000 \\ &= 0.3(25000) - 6000 \\ &= 1500 \end{aligned}$$

Discrete Variables: Measuring Dispersion

- **Variance** of a discrete variable.

$$\sigma^2 = \sum_{i=1}^N [x_i - E(X)]^2 P(X = x_i)$$

- **Standard Deviation** of a discrete variable.

$$\sigma = \sqrt{\sigma^2} = \sqrt{\sum_{i=1}^N [x_i - E(X)]^2 P(X = x_i)}$$

where:

$E(X)$ = Expected value of the discrete variable X

x_i = the i^{th} outcome of X

$P(X=x_i)$ = Probability of the i^{th} occurrence of X

Discrete Variables: Measuring Dispersion

(continued)

$$\sigma = \sqrt{\sum_{i=1}^N [x_i - E(X)]^2 P(X = x_i)}$$

Interruptions Per Day In Computer Network (x_i)	Probability $P(X = x_i)$	$[x_i - E(X)]^2$	$[x_i - E(X)]^2 P(X = x_i)$
0	0.35	$(0 - 1.4)^2 = 1.96$	$(1.96)(0.35) = 0.686$
1	0.25	$(1 - 1.4)^2 = 0.16$	$(0.16)(0.25) = 0.040$
2	0.20	$(2 - 1.4)^2 = 0.36$	$(0.36)(0.20) = 0.072$
3	0.10	$(3 - 1.4)^2 = 2.56$	$(2.56)(0.10) = 0.256$
4	0.05	$(4 - 1.4)^2 = 6.76$	$(6.76)(0.05) = 0.338$
5	0.05	$(5 - 1.4)^2 = 12.96$	$(12.96)(0.05) = 0.648$
			$\sigma^2 = 2.04, \sigma = 1.4283$

Discrete Variables: Measuring Dispersion

- A calculation formula for the **variance**.

$$\sigma^2 = E[X^2] - (E[X])^2 = \sum_{i=1}^N x_i^2 P(X = x_i) - \mu^2$$

where:

$E(X) = \mu$ = Expected value of the discrete variable X

x_i = the i^{th} outcome of X

$P(X=x_i)$ = Probability of the i^{th} occurrence of X

$$\therefore \sigma^2 = E[(X - \mu)^2] = E[X^2] - \mu^2$$

Rules of the Variance

- Rule 1: $Var(c) = 0$
- Rule 2: $Var(X + c) = Var(X)$
- Rule 3: $Var(cX) = c^2 Var(X)$

It therefore follows that, for constants a and b ,

$$Var(aX + b) = a^2 \cdot Var(X)$$

Rules of the Variance:

Example

Monthly sales have a mean of R25 000 and a standard deviation of R4 000. Profits are calculated by multiplying sales by 30% and subtracting fixed costs of R6 000.

*Calculate the **STANDARD DEVIATION** of monthly profit.*

Describe the problem in algebraic terms:

- *Sales have a standard deviation of R4 000*
 $\rightarrow \text{Var}(\text{Sales}) = 4\,000^2 = 16\,000\,000$
- *Profits are calculated by* $\rightarrow \text{Profit} = 0.30(\text{Sales}) - 6\,000$

Rules of the Variance

Let X = the monthly sales. It is given that $Var(X) = 16\,000\,000$.

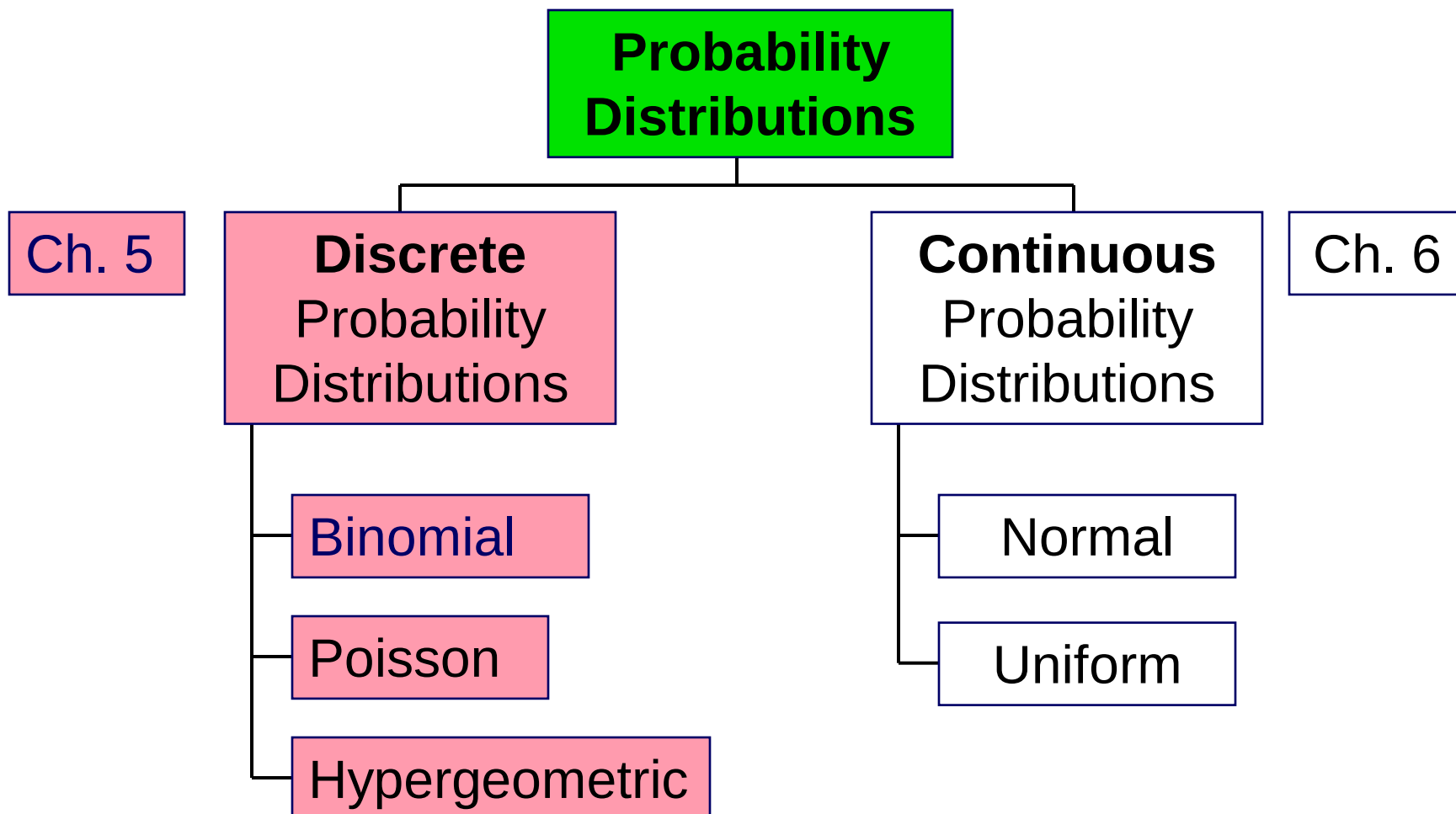
It is given that profit = $P = 0.3X - 6000$.

Hence,

$$\begin{aligned} Var(P) &= Var(0.3X - 6000) \\ &= 0.3^2 Var(X) \\ &= 0.09(16\,000\,000) \\ &= 1\,440\,000 \end{aligned}$$

The **standard deviation** of sales is $\sigma_P = \sqrt{1440000} = 1200$.

Probability Distributions



Binomial Probability Distribution

- A fixed number of observations, n .
 - e.g., 15 tosses of a coin; ten light bulbs taken from a warehouse.
- Each observation is classified into one of two mutually exclusive & collectively exhaustive categories.
 - e.g., head or tail in each toss of a coin; defective or not defective light bulb.
- The probability of being classified as the event of interest, π , is constant from observation to observation.
 - Probability of getting a tail is the same each time we toss the coin.
 - Since the two categories are mutually exclusive and collectively exhaustive, when the probability of the event of interest is π , the probability of the event of interest not occurring is $1 - \pi$.
- The value of any observation is independent of the value of any other observation.

Possible Applications for the Binomial Distribution

- A manufacturing plant labels items as either defective or acceptable.
- A firm bidding for contracts will either get a contract or not.
- A marketing research firm receives survey responses of “yes I will buy” or “no I will not.”
- New job applicants either accept the offer or reject it.

The Binomial Distribution

Counting Techniques

- Suppose the event of interest is obtaining heads on the toss of a fair coin. You are to toss the coin three times. In how many ways can you get two heads?
- Possible ways: HHT, HTH, THH, so there are three ways you can get two heads.
- This situation is fairly simple. We need to be able to count the number of ways for more complicated situations.

Counting Techniques

Rule of Combinations

- The number of combinations of selecting x objects out of n objects is:

$${}_n C_x = \frac{n!}{x!(n-x)!}$$

where:

$$n! = (n)(n-1)(n-2) \cdots (1)$$

$$x! = (x)(x-1)(x-2) \cdots (1)$$

$$0! = 1 \quad (\text{by definition})$$

Counting Techniques

Rule of Combinations

- How many possible 3 scoop combinations could you create at an ice cream parlor if you have 31 flavors to select from and no flavor can be used more than once in the 3 scoops?
- The total choices is $n = 31$, and we select $X = 3$.

$${}_{31}C_3 = \frac{31!}{3!(31-3)!} = \frac{31!}{3!28!} = \frac{31 \bullet 30 \bullet 29 \bullet 28!}{3 \bullet 2 \bullet 1 \bullet 28!} = 31 \bullet 5 \bullet 29 = 4,495$$

Binomial Distribution Formula

$$P(X=x | n, \pi) = \frac{n!}{x! (n-x)!} \pi^x (1-\pi)^{n-x}$$

$P(X=x|n,\pi)$ = probability that $X = x$ events of interest, given n and π

x = number of “events of interest” in sample,
($x = 0, 1, 2, \dots, n$)

n = sample size (number of trials
or observations)

π = probability of “event of interest”

$1 - \pi$ = probability of not having an
event of interest

Example: Flip a coin four times, let x = # heads:

$$n = 4$$

$$\pi = 0.5$$

$$1 - \pi = (1 - 0.5) = 0.5$$

$$X = 0, 1, 2, 3, 4$$

Example: Calculating a Binomial Probability

What is the probability of one success in five observations if the probability of an event of interest is 0.1?

$$x = 1, n = 5, \text{ and } \pi = 0.1$$

$$\begin{aligned} P(X = 1 | 5, 0.1) &= \frac{n!}{x!(n-x)!} \pi^x (1-\pi)^{n-x} \\ &= \frac{5!}{1!(5-1)!} (0.1)^1 (1-0.1)^{5-1} \\ &= (5)(0.1)(0.9)^4 \\ &= 0.32805 \end{aligned}$$

The Binomial Distribution

Example

Suppose the probability of an invoice payment being late is 0.10. What is the probability of 1 late invoice payment in a group of 4 invoices?

$$x = 1, n = 4, \text{ and } \pi = 0.10$$

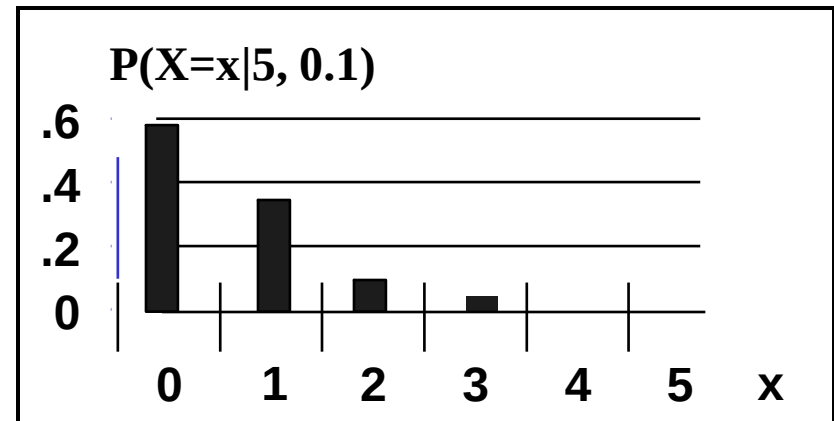
$$\begin{aligned} P(X = 1 | 4, 0.10) &= \frac{n!}{x!(n-x)!} \pi^x (1-\pi)^{n-x} \\ &= \frac{4!}{1!(4-1)!} (0.10)^1 (1-0.10)^{4-1} \\ &= (4)(0.10)(0.729) \\ &= 0.2916 \end{aligned}$$

Excel Can Be Used To Calculate Binomial Probabilities

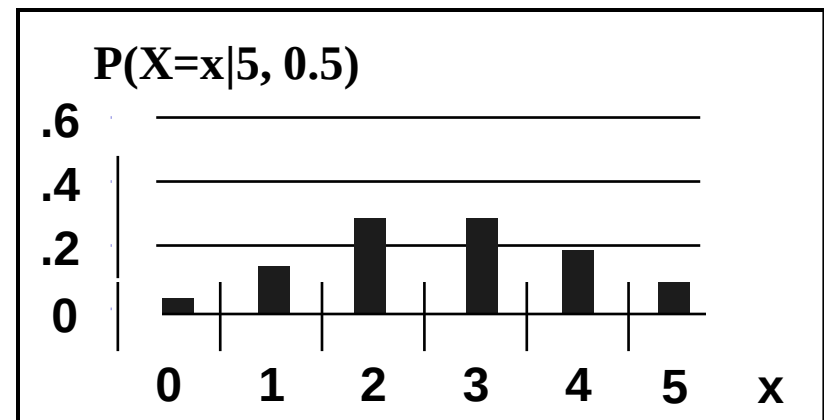
	A	B
1	Binomial Probabilities	
2		
3	Data	
4	Sample size	4
5	Probability of an event of interest	0.1
6		
7	Parameters	
8	Mean	0.4 =B4 * B5
9	Variance	0.36 =B8 * (1 - B5)
10	Standard deviation	0.6 =SQRT(B9)
11		
12	Binomial Probabilities Table	
13	X	P(X)
14	0	0.6561 =BINOM.DIST(A14, \$B\$4, \$B\$5, FALSE)
15	1	0.2916 =BINOM.DIST(A15, \$B\$4, \$B\$5, FALSE)
16	2	0.0486 =BINOM.DIST(A16, \$B\$4, \$B\$5, FALSE)
17	3	0.0036 =BINOM.DIST(A17, \$B\$4, \$B\$5, FALSE)
18	4	0.0001 =BINOM.DIST(A18, \$B\$4, \$B\$5, FALSE)

The Binomial Distribution Shape

- The shape of the binomial distribution depends on the values of π and n .
- Here, $n = 5$ and $\pi = 0.1$.



- Here, $n = 5$ and $\pi = 0.5$.



The Binomial Distribution Characteristics

- Mean:

$$\mu = E(X) = n\pi$$

- Variance and Standard Deviation:

$$\sigma^2 = n\pi(1 - \pi)$$

$$\sigma = \sqrt{n\pi(1 - \pi)}$$

Where n = sample size

π = probability of the event of interest for any trial

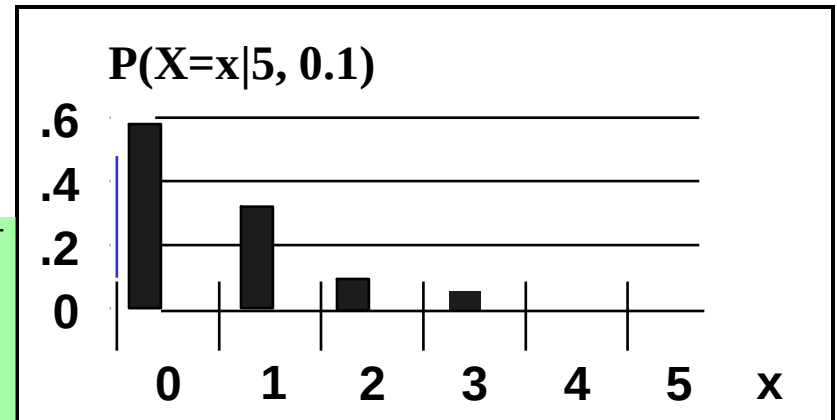
$(1 - \pi)$ = probability of no event of interest for any trial

The Binomial Distribution Characteristics

Examples

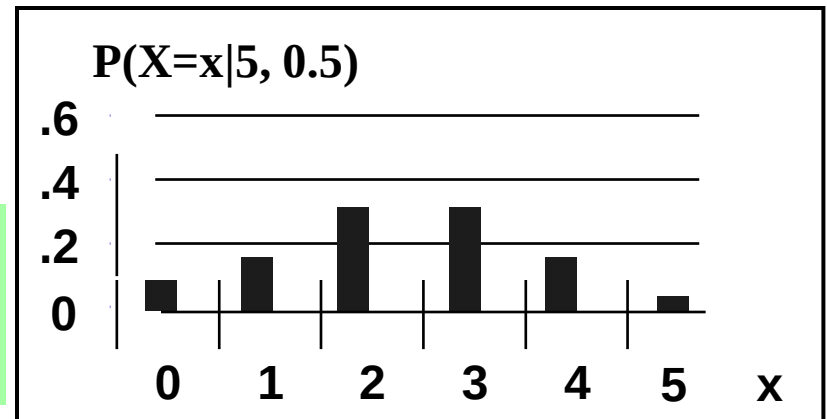
$$\mu = n\pi = (5)(.1) = 0.5$$

$$\sigma = \sqrt{n\pi(1 - \pi)} = \sqrt{(5)(.1)(1 - .1)} \\ = 0.6708$$



$$\mu = n\pi = (5)(.5) = 2.5$$

$$\sigma = \sqrt{n\pi(1 - \pi)} = \sqrt{(5)(.5)(1 - .5)} \\ = 1.118$$



Cumulative Probabilities

Thus far, we have been using a discrete probability distribution to find probabilities for individual values of X . A **cumulative probability** considers the probability that X is **less than or equal** to x , that is $P(X \leq x)$.

For example, compute $P(X \leq 3)$ or $P(X \geq 1)$.

Cumulative Probabilities

The **binomial table** gives cumulative probabilities for $P(X \leq k)$. This table can be used to compute **marginal** and **excess** probabilities:

$$P(X = k) = P(X \leq k) - P(X \leq [k-1]).$$

Likewise, for probabilities given as $P(X \geq k)$, we have:

$$P(X \geq k) = 1 - P(X \leq [k-1]).$$

The Poisson Distribution

Definitions

- You use the **Poisson distribution** when you are interested in the number of times an event occurs in a given **area of opportunity**.
- An **area of opportunity** is a continuous unit or interval of time, volume, or such area in which more than one occurrence of an event can occur.
 - The number of scratches in a car's paint.
 - The number of mosquito bites on a person.
 - The number of computer crashes in a day

The Poisson Distribution

- Apply the Poisson Distribution when:
 - You are interested in counting the number of times a particular event occurs in a given area of opportunity. An area of opportunity is defined by time, length, surface area, and so forth.
 - The probability that an event occurs in a given area of opportunity is the same for all the areas of opportunity.
 - The number of events that occur in one area of opportunity is independent of the number of events that occur in any other area of opportunity.
 - The probability that two or more events will occur in an area of opportunity approaches zero as the area of opportunity becomes smaller.
- The average number of events per unit is λ (lambda).

The Poisson Distribution Formula

$$P(X = x | \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$$

where:

x = number of events in an area of opportunity

λ = expected number of events

e = base of the natural logarithm system (2.71828...)

The Poisson Distribution Characteristics

- Mean:

$$\mu = \lambda$$

- Variance and Standard Deviation:

$$\sigma^2 = \lambda$$

$$\sigma = \sqrt{\lambda}$$

where λ = expected number of events.

The Poisson Distribution Example

The mean number of customers who arrive per minute at a bank during the noon-to-1pm hour is 3.0. What is the probability that 2 customers arrive in a given minute?

$$x = 2, \lambda = 3.0$$

$$\begin{aligned} P(X = 2|3.0) &= \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-3.0} 3.0^2}{2!} \\ &= \frac{9}{2.71828^3 (2)} \\ &= 0.2240 \end{aligned}$$

Excel Can Automate Poisson Probability Calculations

The mean number of customers who arrive per minute at a bank during the noon-to-1pm hour is 3.0.

$$\lambda = 3.0$$

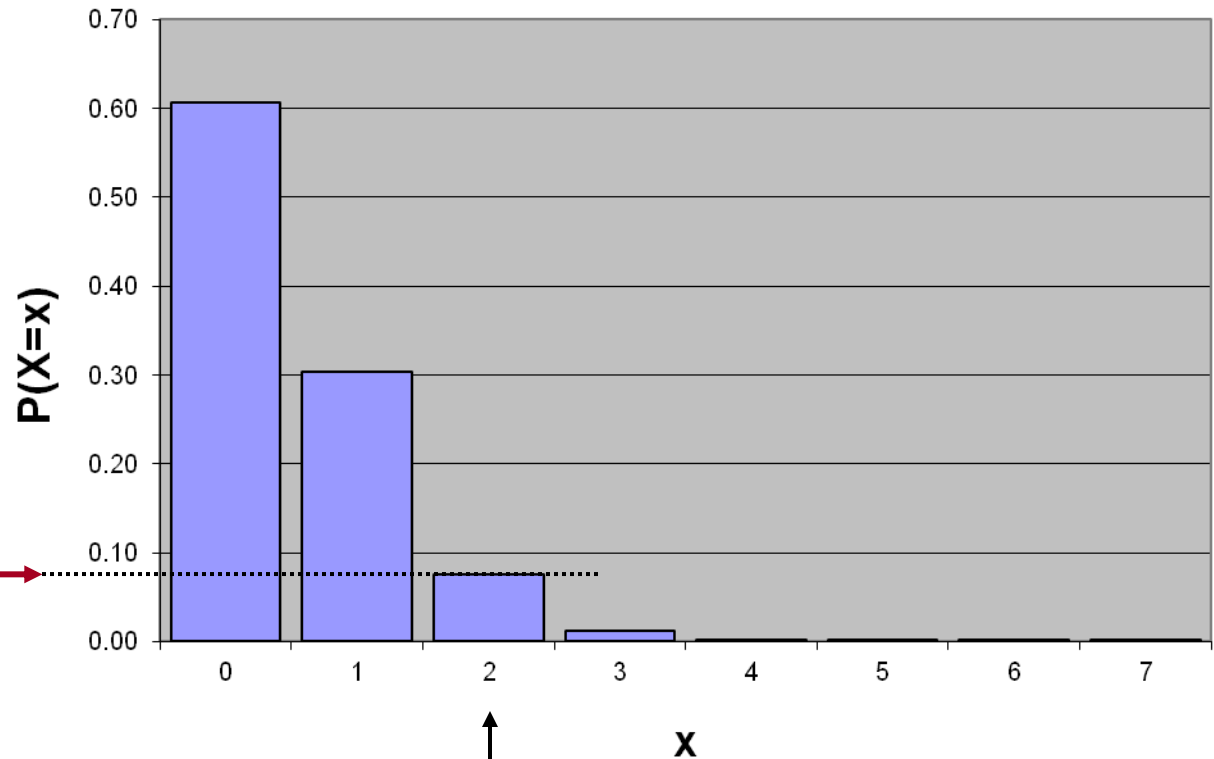
	A	B	C	D	E
1	Poisson Probabilities				
2					
3	Data				
4	Mean/Expected number of events of interest:				3
5					
6	Poisson Probabilities Table				
7	X	P(X)			
8	0	0.0498	=POISSON.DIST(A8, \$E\$4, FALSE)		
9	1	0.1494	=POISSON.DIST(A9, \$E\$4, FALSE)		
10	2	0.2240	=POISSON.DIST(A10, \$E\$4, FALSE)		
11	3	0.2240	=POISSON.DIST(A11, \$E\$4, FALSE)		
12	4	0.1680	=POISSON.DIST(A12, \$E\$4, FALSE)		
13	5	0.1008	=POISSON.DIST(A13, \$E\$4, FALSE)		
14	6	0.0504	=POISSON.DIST(A14, \$E\$4, FALSE)		
15	7	0.0216	=POISSON.DIST(A15, \$E\$4, FALSE)		
16	8	0.0081	=POISSON.DIST(A16, \$E\$4, FALSE)		
17	9	0.0027	=POISSON.DIST(A17, \$E\$4, FALSE)		
18	10	0.0008	=POISSON.DIST(A18, \$E\$4, FALSE)		
19	11	0.0002	=POISSON.DIST(A19, \$E\$4, FALSE)		
20	12	0.0001	=POISSON.DIST(A20, \$E\$4, FALSE)		
21	13	0.0000	=POISSON.DIST(A21, \$E\$4, FALSE)		
22	14	0.0000	=POISSON.DIST(A22, \$E\$4, FALSE)		
23	15	0.0000	=POISSON.DIST(A23, \$E\$4, FALSE)		

Graph of Poisson Probabilities

Graphically:

$\lambda = 0.50$

X	$\lambda = 0.50$
0	0.6065
1	0.3033
2	0.0758
3	0.0126
4	0.0016
5	0.0002
6	0.0000
7	0.0000

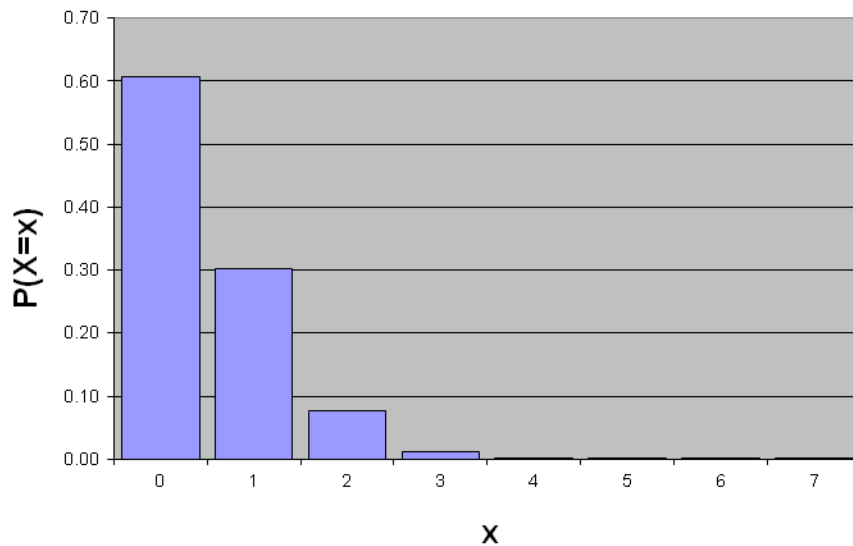


$$P(X = 2 \mid \lambda=0.50) = 0.0758$$

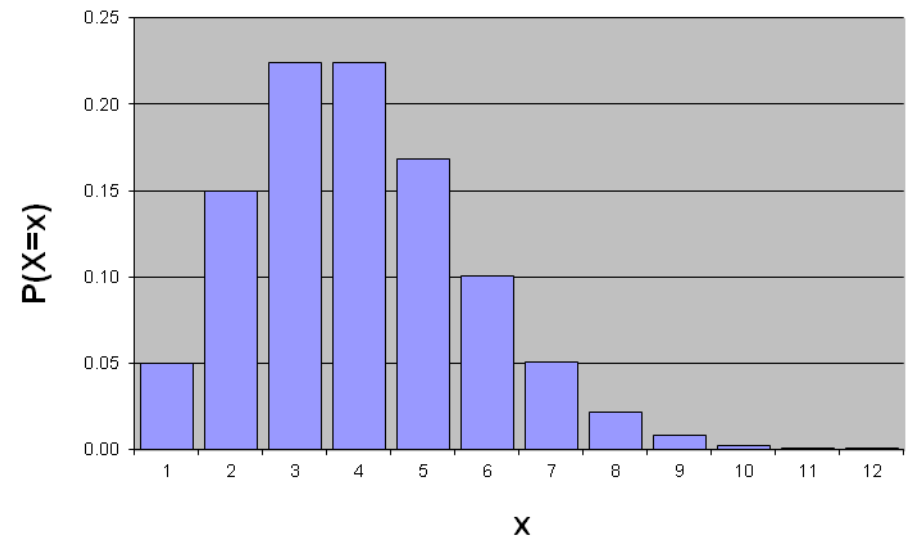
Poisson Distribution Shape

- The shape of the Poisson Distribution depends on the parameter λ :

$$\lambda = 0.50$$



$$\lambda = 3.00$$



The Hypergeometric Distribution

- The **binomial distribution** is applicable when selecting from a finite population with replacement or from an infinite population without replacement.
- The **hypergeometric distribution** is applicable when selecting from a finite population without replacement.

The Hypergeometric Distribution

- “n” trials in a sample taken from a **finite population** of size N.
- Sample taken **without replacement**.
- Outcomes of trials are **dependent**.
- Concerned with finding the probability of “X” items of interest in the sample where there are “E” items of interest in the population.

The Hypergeometric Distribution Formula

$$P(X = x \mid n, N, E) = \frac{\binom{E}{x} \binom{N-E}{n-x}}{\binom{N}{n}}$$

Where

N = population size

E = number of items of interest in the population

$N - E$ = number of events not of interest in the population

n = sample size

x = number of items of interest in the sample

$n - x$ = number of events not of interest in the sample

Properties of the Hypergeometric Distribution

- The **mean** of the hypergeometric distribution is:

$$\mu = E(X) = \frac{nE}{N}$$

- The **standard deviation** is:

$$\sigma = \sqrt{\frac{nE(N-E)}{N^2}} \cdot \sqrt{\frac{N-n}{N-1}}$$

Where $\sqrt{\frac{N-n}{N-1}}$ is called the “**Finite Population Correction Factor**” from sampling without replacement from a finite population

Using the Hypergeometric Distribution

8 different computers are checked out from 30 in the department. 10 of the 30 computers have illegal software loaded. What is the probability that 2 of the 8 selected computers have illegal software loaded?

$N = 30$	$n = 8$
$E = 10$	$x = 2$

$$P(X = 2|8,30,10) = \frac{\binom{E}{x} \binom{N-E}{n-x}}{\binom{N}{n}} = \frac{\binom{10}{2} \binom{20}{6}}{\binom{30}{8}} = \frac{(45)(38,760)}{5,852,925} = 0.2980$$

The probability that 2 of the 8 selected computers have illegal software loaded is 0.2980, or 29.8%.

Excel Can Be Used For The Hypergeometric Distribution

8 different computers are checked out from 30 in the department. 10 of the 30 computers have illegal software loaded. What is the probability that 2 of the 8 selected computers have illegal software loaded?

$$N = 30$$

$$E = 10$$

$$n = 8$$

$$x = 2$$

	A	B
1	Hypergeometric Probabilities	
2		
3	Data	
4	Sample size	8
5	No. of events of interest in population	10
6	Population size	30
7		
8	Hypergeometric Probabilities Table	
9	X	P(X)
10	0	0.0215
11	1	0.1324
12	2	0.2980
13	3	0.3179
14	4	0.1738
15	5	0.0491
16	6	0.0068
17	7	0.0004
18	8	0.0000

=HYPGEOM.DIST(A10, \$B\$4, \$B\$5, \$B\$6, FALSE)

=HYPGEOM.DIST(A11, \$B\$4, \$B\$5, \$B\$6, FALSE)

=HYPGEOM.DIST(A12, \$B\$4, \$B\$5, \$B\$6, FALSE)

=HYPGEOM.DIST(A13, \$B\$4, \$B\$5, \$B\$6, FALSE)

=HYPGEOM.DIST(A14, \$B\$4, \$B\$5, \$B\$6, FALSE)

=HYPGEOM.DIST(A15, \$B\$4, \$B\$5, \$B\$6, FALSE)

=HYPGEOM.DIST(A16, \$B\$4, \$B\$5, \$B\$6, FALSE)

=HYPGEOM.DIST(A17, \$B\$4, \$B\$5, \$B\$6, FALSE)

=HYPGEOM.DIST(A18, \$B\$4, \$B\$5, \$B\$6, FALSE)

Discrete Bivariate Distributions

Until now, we have considered *univariate distributions*, i.e. probability distributions in **ONE** variable.

Bivariate distributions are probabilities of combinations of **TWO** variables.

Bivariate probability distributions are also called joint probabilities. A **joint probability distribution** of X and Y is a table or formula that lists the joint probabilities for all **PAIRS** of values x and y , and is denoted $P(x, y)$.

$$P(x, y) = P(X = x \text{ and } Y = y)$$

Discrete Bivariate Distributions

The requirements for a bivariate distribution are like a univariate distribution, with only minor changes to the notation:

1. $0 \leq P(x, y) \leq 1$
2. $\sum_{all\ x} \sum_{all\ y} P(x, y) = 1$
for all pairs (x, y) .

Discrete Bivariate Distributions

Example

Xavier and Yvette are real estate agents. Let X denote the number of houses that Xavier will sell in a month and let Y denote the number of houses Yvette will sell in a month. An analysis of their past monthly performances has the following joint probabilities.

		X			
		0	1	2	
Y	0	0.12	0.42	0.06	0.60
	1	0.21	0.06	0.03	0.30
	2	0.07	0.02	0.01	0.10
		0.40	0.50	0.10	1.00

Discrete Bivariate Distributions

Marginal Probabilities

We can calculate the **marginal probabilities** by summing across rows and down columns to determine the probabilities of X and Y individually:

		X			
		0	1	2	
Y	0	0.12	0.42	0.06	0.60
	1	0.21	0.06	0.03	0.30
	2	0.07	0.02	0.01	0.10
		0.40	0.50	0.10	1.00

X	$P(x)$
0	0.4
1	0.5
2	0.1

Y	$P(y)$
0	0.6
1	0.3
2	0.1

Discrete Bivariate Distributions

We can describe the mean, variance, and standard deviation of *each variable* in a bivariate distribution by working with the *marginal probabilities*.

X	$P(x)$
0	0.4
1	0.5
2	0.1

$$E(X) = \sum_{i=1}^3 x_i P(X_i) = 0(0.4) + 1(0.5) + 2(0.1) = 0.7$$

$$\begin{aligned} \text{Var}(X) &= \sigma^2 = E(X^2) - [E(X)]^2 \\ &= 0^2(0.4) + 1^2(0.5) + 2^2(0.1) - [0.7]^2 \\ &= 0.41 \end{aligned}$$

$$\sigma = \sqrt{\sigma^2} = \sqrt{0.41} = 0.64$$

Discrete Bivariate Distributions

We can describe the mean, variance, and standard deviation of *each variable* in a bivariate distribution by working with the *marginal probabilities*.

Y	$P(y)$
0	0.6
1	0.3
2	0.1

$$E(Y) = \sum_{j=1}^3 y_j P(Y_j) = 0(0.6) + 1(0.3) + 2(0.1) = 0.5$$

$$\begin{aligned} \text{Var}(Y) &= \sigma^2 = E(Y^2) - [E(Y)]^2 \\ &= 0^2(0.6) + 1^2(0.3) + 2^2(0.1) - [0.5]^2 \\ &= 0.45 \end{aligned}$$

$$\sigma = \sqrt{\sigma^2} = \sqrt{0.45} = 0.671$$

Covariance

- The covariance measures the strength of the linear relationship between two discrete variables X and Y .
- A positive covariance indicates a positive relationship.
- A negative covariance indicates a negative relationship.

The Covariance Formula

- The **covariance** formula:

$$\sigma_{XY} = \sum_{i=1}^N [x_i - E(X)][y_i - E(Y)] P(X = x_i, Y = y_i)$$

where: X = discrete variable X

x_i = the i^{th} outcome of X

Y = discrete variable Y

y_i = the i^{th} outcome of Y

$P(X=x_i, Y=y_i)$ = probability of occurrence of the
 i^{th} outcome of X and the i^{th} outcome of Y

The Covariance Formula

The **covariance** has a calculation formula:

$$COV(X, Y) = \sigma_{xy} = \sum_{all\ x} \sum_{all\ y} x_i y_j P(x_i, y_j) - \mu_X \mu_Y$$

which is equivalent to:

$$COV(X, Y) = \sigma_{xy} = E(XY) - E(X)E(Y)$$



Coefficient of Correlation

The coefficient of correlation is calculated and interpreted in the same way as described in Chapter 3:

$$\rho = \frac{COV(X, Y)}{\sigma_X \sigma_Y}$$

See Chapter 3 slides 64 - 66 for the interpretation.

Covariance: Example

Compute the *covariance* and the *coefficient of correlation* between the numbers of houses sold by Xavier and Yvette.

$$\begin{aligned} COV(X, Y) &= \sigma_{xy} = \sum_{\text{all } x} \sum_{\text{all } y} x_i y_j P(x_i, y_j) - \mu_X \mu_Y \\ &= 0 \times 0 \times (0.12) + 0 \times 1 \times (0.21) + 0 \times 2 \times (0.07) + 1 \times 0 \\ &\times (0.42) + 1 \times 1 \times (0.06) + 1 \times 2 \times (0.02) + 2 \times 0 \times (0.06) \\ &+ 2 \times 1 \times (0.03) + 2 \times 2 \times (0.01) - (0.7)(0.5) \\ &= 0.2 - 0.35 = -0.15 \end{aligned}$$

Covariance: Example

$$\rho = \frac{COV(X, Y)}{\sigma_X \sigma_Y} = \frac{-0.150}{(0.64)(0.671)} = -0.35$$

Interpretation:

There is a moderate, negative linear relationship between the number of houses sold by Xavier and the number of houses sold by Yvette.

Sum of Two Random Variables

A bivariate distribution allows us to develop the probability distribution of *any* combination of the two variables, of particular interest is the **SUM** of two variables.

If we consider our example of Xavier and Yvette selling houses, we can create a probability distribution to answer questions like “what is the probability that two houses are sold”?

Sum of Two Random Variables

		X			
		0	1	2	
Y	0	0.12	0.42	0.06	0.60
	1	0.21	0.06	0.03	0.30
	2	0.07	0.02	0.01	0.10
		0.40	0.50	0.10	1.00

X+Y	0	1	2	3	4
P(X+Y)	0.12	0.63	0.19	0.05	0.01

For example:

$$P(X+Y=2) =$$

$$P(0,2) + P(1,1) + P(2,0) = 0.07 + 0.06 + 0.06 = 0.19$$

Sum of Two Random Variables

We can compute the expected value of $X+Y$ in the usual way.

$X+Y$	0	1	2	3	4
$P(X+Y)$	0.12	0.63	0.19	0.05	0.01

$$\begin{aligned}E(X + Y) &= 0(0.12) + 1(0.63) + 2(0.19) + 3(0.05) + 4(0.01) \\&= 1.2\end{aligned}$$

Sum of Two Random Variables

Likewise, we can compute the variance and standard deviation of $X+Y$ in the usual way.

$X+Y$	0	1	2	3	4
$P(X+Y)$	0.12	0.63	0.19	0.05	0.01

$$\begin{aligned} \text{Var}(X + Y) &= E(X + Y)^2 - [E(X + Y)]^2 = \\ &0^2(0.12) + 1^2(0.63) + 2^2(0.19) + 3^2(0.05) + 4^2(0.01) - (1.2)^2 \\ &= 0.56 \end{aligned}$$

$$\sigma_{X+Y} = \sqrt{\text{Var}(X + Y)} = \sqrt{0.56} = 0.748$$

Independence

If two random variables are independent, the covariance is equal to zero:

$$COV(X, Y) = 0$$

Then the coefficient of correlation, ρ , is also equal to zero:

$$\rho = \frac{COV(X, Y)}{\sigma_X \sigma_Y} = 0$$

Rules For The Sum Of Two Random Variables

We can derive rules of expected value and variance for the sum of two variables as follows:

- $E(X + Y) = E(X) + E(Y)$
- $V(X + Y) = V(X) + V(Y) + 2COV(X, Y)$

If X and Y are independent, $COV(X, Y) = 0$ and thus:

- $V(X + Y) = V(X) + V(Y)$

Rules For The Sum Of Two Random Variables

$$E(X + Y) = E(X) + E(Y) = 0.7 + 0.5 = 1.2$$

$$\begin{aligned} V(X + Y) &= V(X) + V(Y) + 2COV(X, Y) \\ &= 0.41 + 0.45 + 2(-0.15) = 0.56 \end{aligned}$$

NOTE:

$$E[aX + bY] = aE[X] + bE[Y]$$

$$V(aX + bY) = a^2V(X) + b^2V(Y) + 2abCOV(X, Y)$$

Applications in Finance

Computing the Mean for Investment Returns

Return per R1 000 for two types of investments

$P(x_i, y_i)$	Economic Condition	Investment	
		Passive Fund X	Aggressive Fund Y
0.2	Recession	- 25	- 200
0.5	Stable Economy	+ 50	+ 60
0.3	Expanding Economy	+ 100	+ 350

$$E(X) = \mu_X = (-25)(0.2) + (50)(0.5) + (100)(0.3) = 50$$

$$E(Y) = \mu_Y = (-200)(0.2) + (60)(0.5) + (350)(0.3) = 95$$

Applications in Finance

Computing the Standard Deviation for Investment Returns

$P(x_i, y_i)$	Economic Condition	Investment	
		Passive Fund X	Aggressive Fund Y
0.2	Recession	- 25	- 200
0.5	Stable Economy	+ 50	+ 60
0.3	Expanding Economy	+ 100	+ 350

$$\sigma_X = \sqrt{E(X^2) - [E(X)]^2} = \sqrt{4375 - (50)^2} = 43.301$$

$$\sigma_Y = \sqrt{E(Y^2) - [E(Y)]^2} = \sqrt{46550 - (95)^2} = 193.714$$

Applications in Finance

Computing the Covariance for Investment Returns

$P(x_i, y_i)$	Economic Condition	Investment	
		Passive Fund X	Aggressive Fund Y
0.2	Recession	- 25	- 200
0.5	Stable Economy	+ 50	+ 60
0.3	Expanding Economy	+ 100	+ 350

$$\begin{aligned}
 \sigma_{XY} &= COV(X, Y) = \sigma_{xy} = \sum_{\text{all } x} \sum_{\text{all } y} x_i y_j P(x_i, y_j) - \mu_X \mu_Y \\
 &= (-25)(-200)(0.2) + (50)(60)(0.5) + (100)(350)(0.3) - (50)(95) \\
 &= 13000 - 4750 = 8250
 \end{aligned}$$

Applications in Finance

Interpreting the Results for Investment Returns

The aggressive fund has a higher expected return, but much more risk

$$\mu_Y = 95 > \mu_X = 50$$

BUT

$$\sigma_Y = 193.71 > \sigma_X = 43.30$$

The **COVARIANCE** of 8250 indicates that the two investments are positively related.

Applications in Finance

Portfolio Risk and Return

Portfolio expected return (weighted average return):

$$E(P) = wE(X) + (1 - w)E(Y)$$

Portfolio risk (weighted variability):

$$\sigma_p = \sqrt{w^2\sigma_X^2 + (1 - w)^2\sigma_Y^2 + 2w(1 - w)\sigma_{xy}}$$

where

w = portion of portfolio value in asset X

$(1 - w)$ = portion of portfolio value in asset Y

Example

Investment X:	$\mu_X = 50$	$\sigma_X = 43.30$
Investment Y:	$\mu_Y = 95$	$\sigma_Y = 193.21$
		$\sigma_{XY} = 8250$

Suppose 40% of the portfolio is invested in Investment X and 60% is in Investment Y:


$$E(P) = 0.4(50) + 0.6(95) = 77$$

Example

$$\begin{array}{llll} \text{Investment X:} & \mu_X & = & 50 \quad \sigma_X = 43.30 \\ \text{Investment Y:} & \mu_Y & = & 95 \quad \sigma_Y = 193.21 \\ & & & \sigma_{XY} = 8250 \end{array}$$

Suppose 40% of the portfolio is in Investment X and 60% is in Investment Y:

$$\begin{aligned} \sigma_P &= \sqrt{(0.4)^2 (43.30)^2 + (0.6)^2 (193.71)^2 + 2(0.4)(0.6)(8250)} \\ &= 133.30 \end{aligned}$$

The portfolio return and portfolio variability are between the values for investments X and Y considered individually.

Chapter Summary

In this chapter we covered:

- The properties of a probability distribution.
- Computing the expected value and variance of a probability distribution.
- Computing probabilities from binomial, Poisson and hypergeometric distributions.
- Using the binomial, Poisson, or hypergeometric distributions to solve business problems.
- Applying bivariate distribution to investment management.